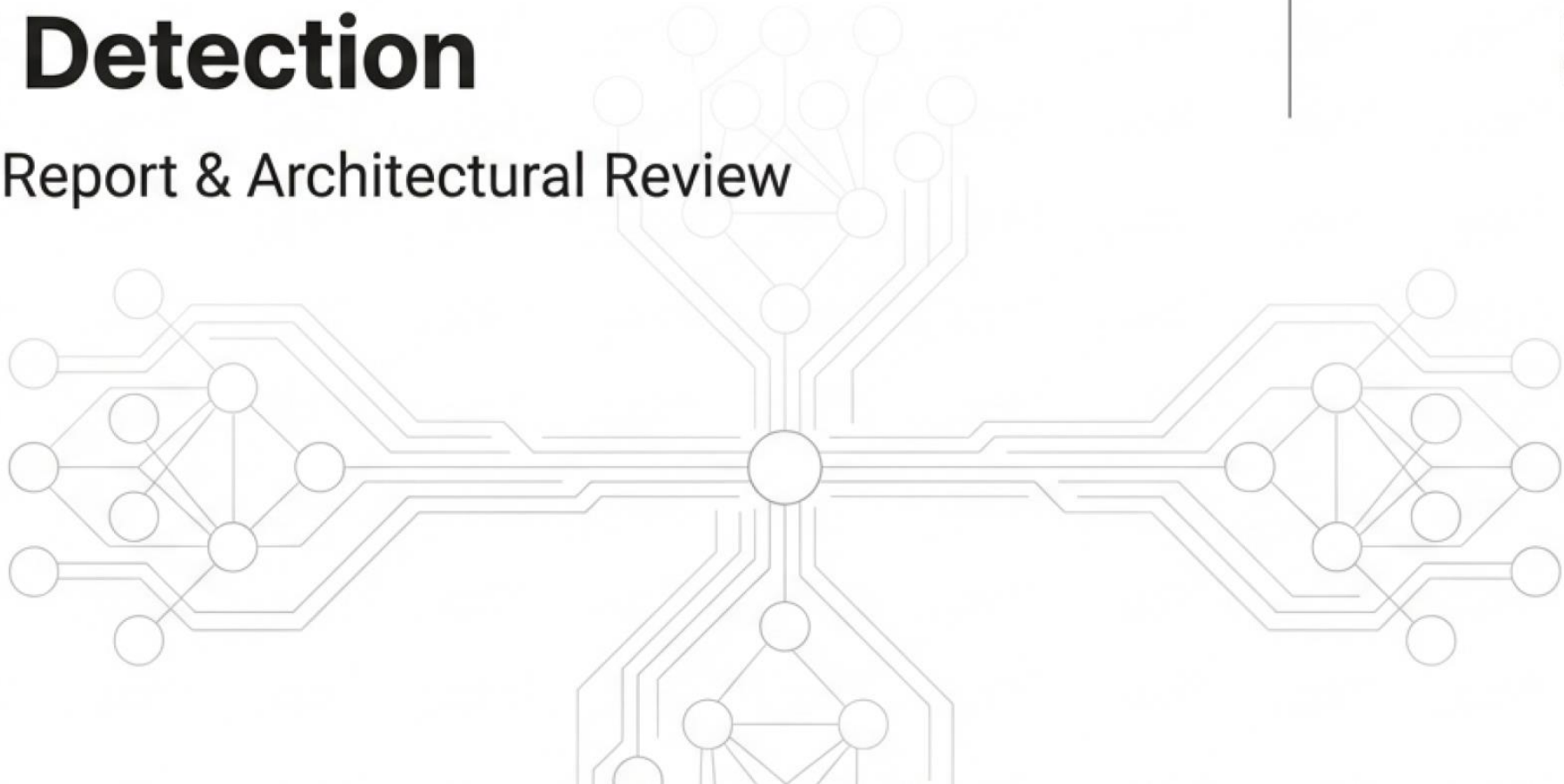
Agentic AI Pub/Sub Infrastructure: A Decentralized Architecture for Real-Time Fraud Detection

Technical Report & Architectural Review

COURSE:	CSYE 7380
PRESENTER:	Renna Wu
DATE:	Jan 31, 2026
STACK:	Apache Kafka, Docker Compose, Python 3.10+, Kafka-Python Client



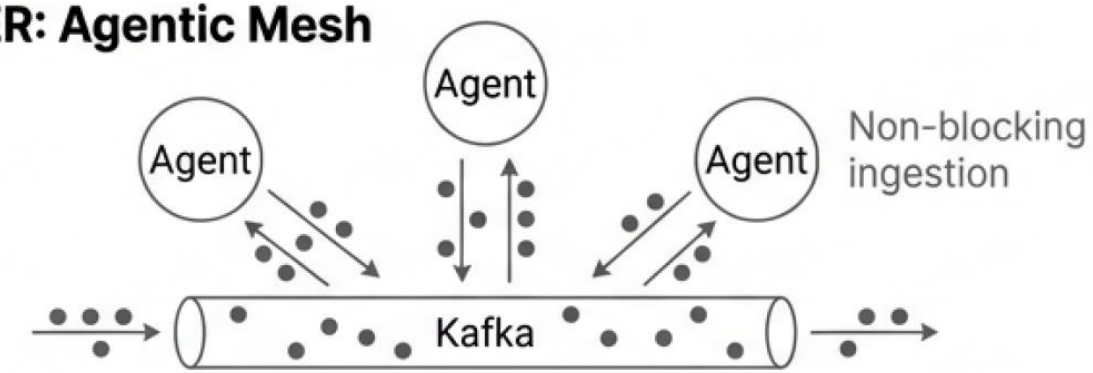
Decentralized architectures replace monolithic latency with event-driven agility

ARCHITECTURAL SHIFT

BEFORE: Monolithic



AFTER: Agentic Mesh

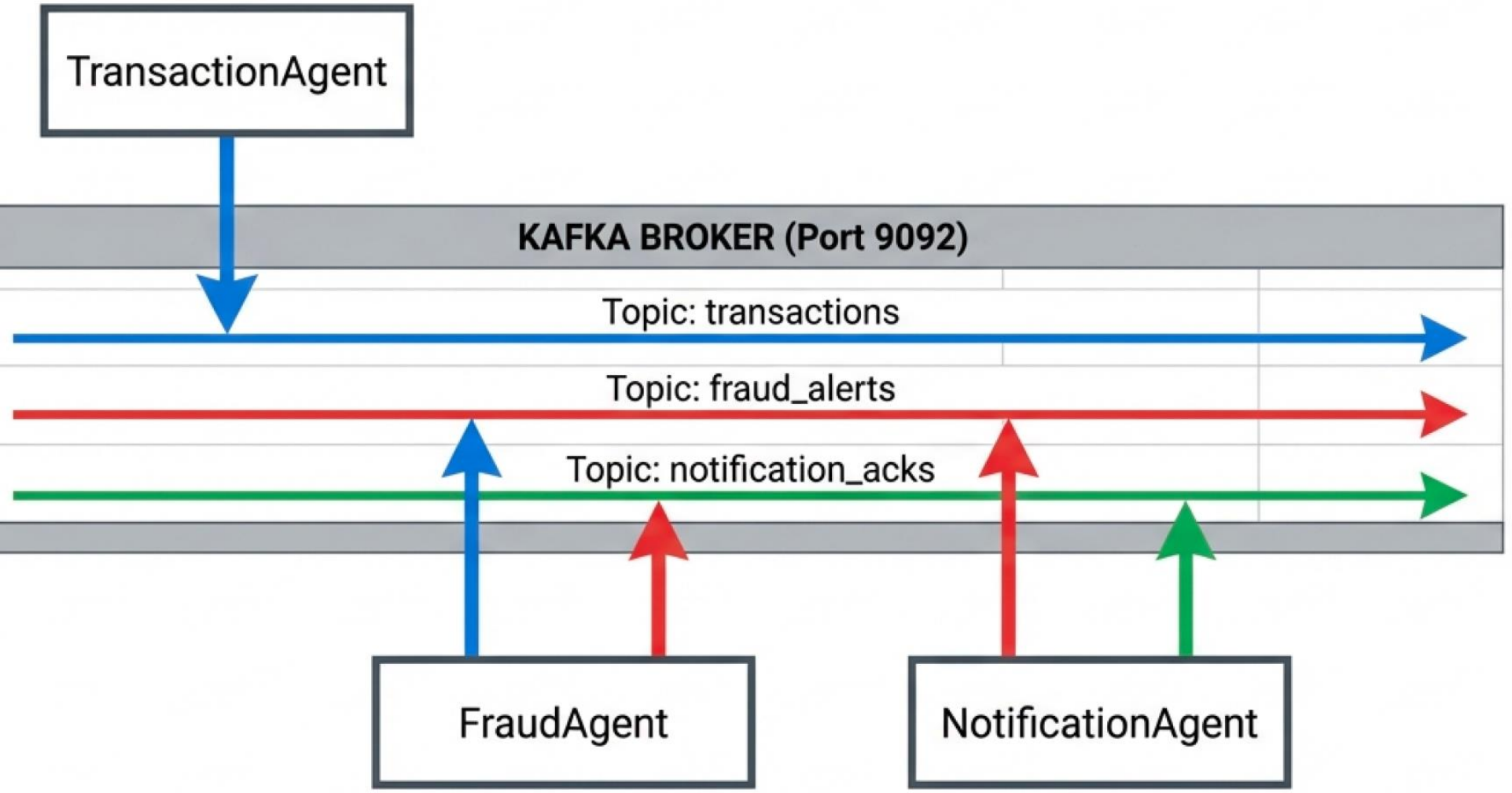


PROJECT SCOPE

This project implements a decentralized Transaction → Fraud → Notification pipeline using Kafka's Pub/Sub pattern. Unlike monolithic architectures, this approach enables independent components to react to environmental messages.

KEY INSIGHT
The core benefit is the separation of ingestion from analysis. The system facilitates real-time fraud detection and automated feedback loops without blocking the primary transaction stream.

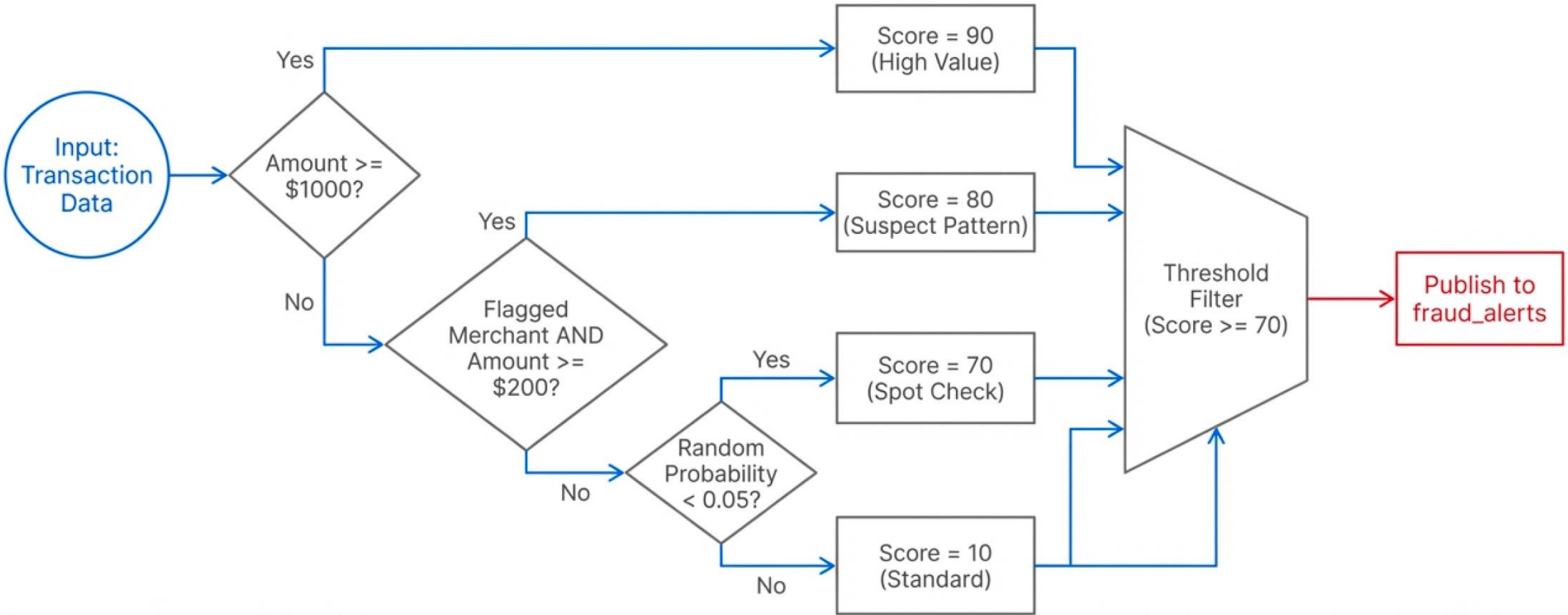
Kafka topics act as a central nervous system, enabling zero-dependency agent interaction.



ARCHITECTURAL SPECS

- Decoupling: Agents interact solely with the broker; no direct service discovery required.
- Asynchronous Processing: Ingestion continues regardless of downstream analysis speed.
- Persistence: Message offsets allow seamless resumption after crashes.

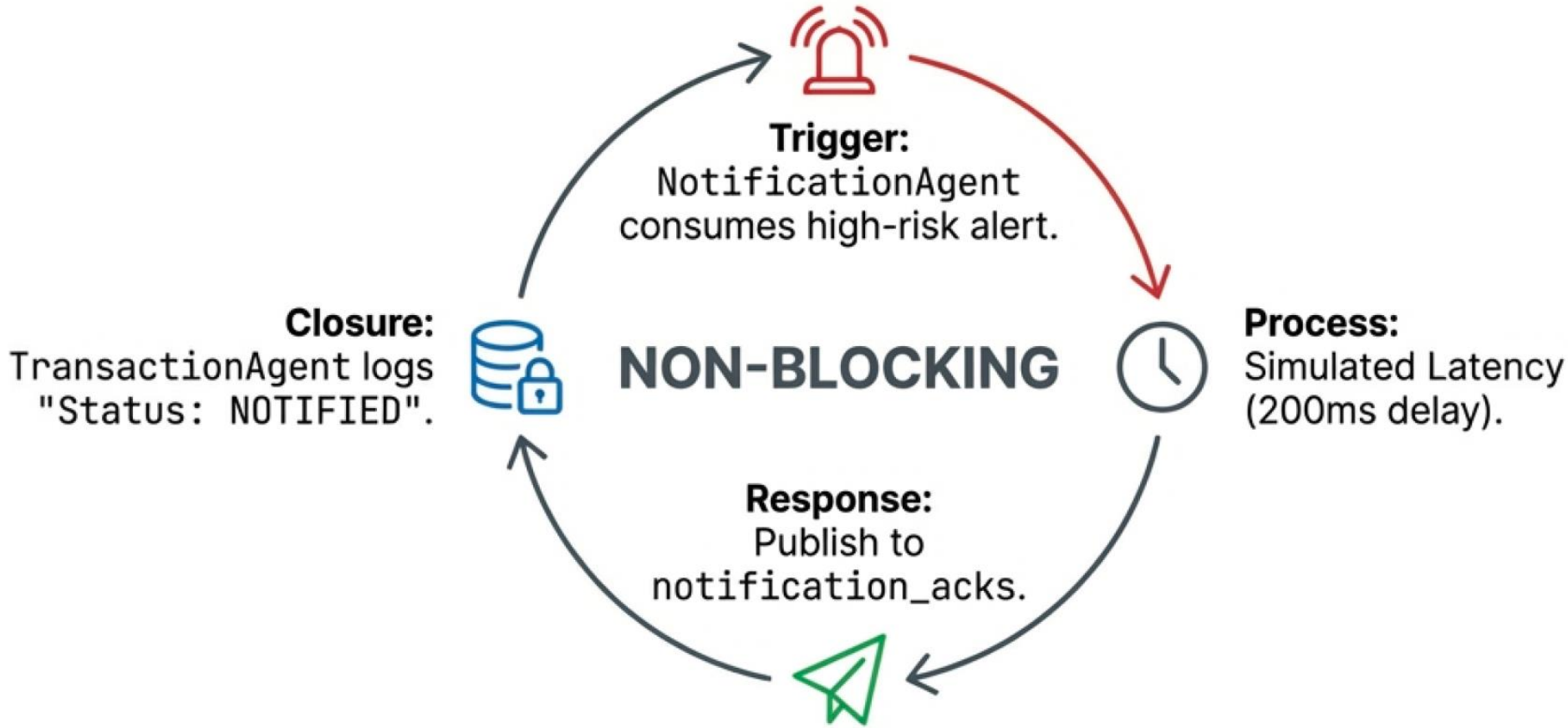
Deterministic scoring logic filters high-volume noise into actionable risk signals.



Implementation: Python 3.10+
Standalone Process

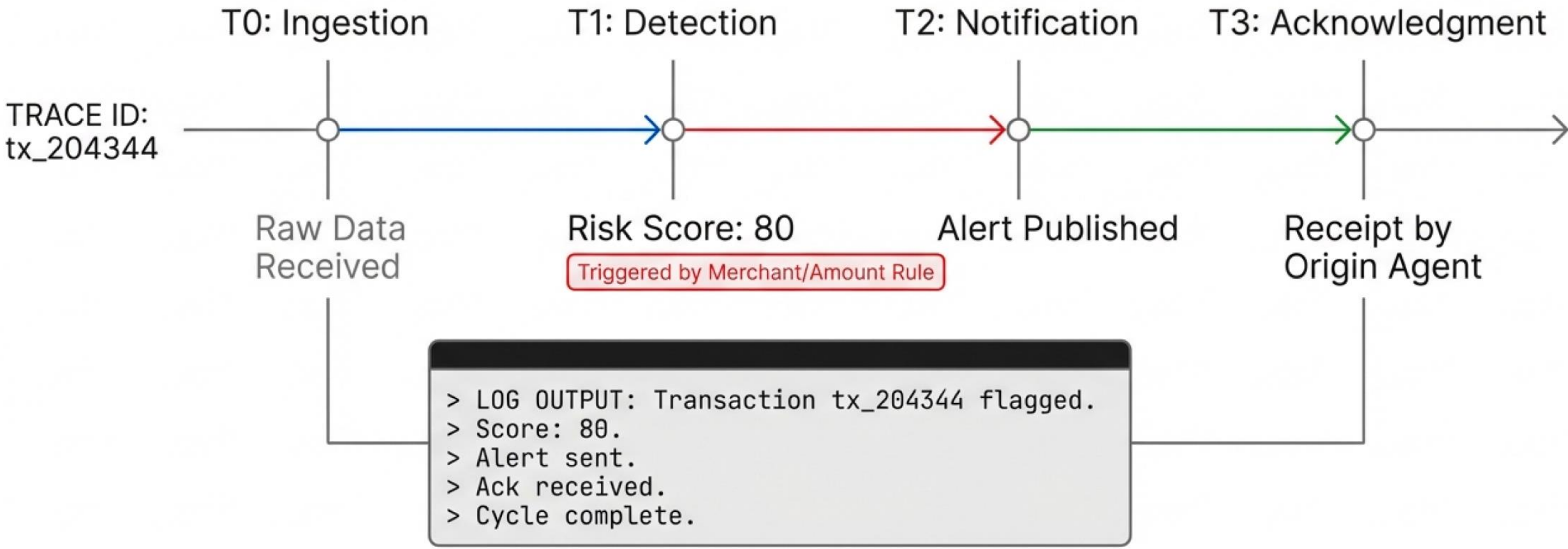
Heuristic rules prioritize financial risk over random sampling.

Asynchronous feedback loops maintain system state without blocking ingestion.



The **TransactionAgent** does not wait for the ack to process the next incoming transaction. It receives the ack only when the loop closes, ensuring high throughput.

Lifecycle verification confirms real-time detection and response for high-risk payloads.



Containerized infrastructure ensures persistence and rapid recovery from failure states.

System Health

CONTAINER STATUS

kafka_broker: OPERATIONAL



CONNECTIVITY

Active Port: 9092



STABILITY METRICS

Uptime: 41 Minutes

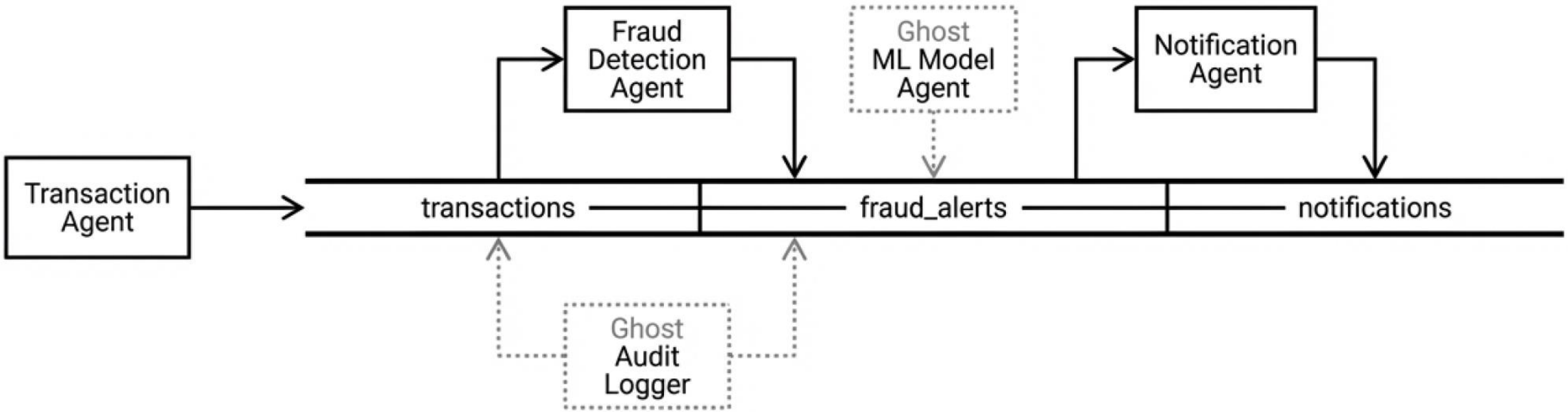
Stable during test window.

RESILIENCE PROTOCOL

Offset Retention Active

Agents can crash and restart without data loss. Message offsets are stored on disk, allowing playback from point of failure.

The pub/sub foundation supports linear scalability and transparent audit trails.



Linear Scalability

Increase volume handling simply by adding more agents to consumer groups. The bus handles the load distribution.

Auditability

Every decision is transparently logged. Reasoning and scoring are embedded in the event payload history.

Extensibility

Scoring models can be swapped (e.g., changing the $1000 threshold) without taking down the pipeline or re-engineering architecture.